TECHNICAL DOCUMENTATION

PDFolio ? Technical & Architecture Guide

Complete specification of system architecture, hardware requirements, CLI engines, zero-retention storage security, and production deployment.

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STACK:	Laravel 12 (PHP 8.3) | Vue.js 3 | Tailwind CSS 4
ENGINES:	QPDF, Ghostscript, Poppler-utils, LibreOffice, OCRmyPDF, FPDI
VERSION:	2.0 Production Ready (October 2026)

1. Architectural Overview & Philosophy

PDFolio is an enterprise-grade, self-hosted web application engineered to replace third-party cloud PDF utilities. Designed for institutions handling sensitive corporate, legal, and financial files, all document processing is executed entirely within your own infrastructure.

- **Stateless Request Architecture:** The server retains no customer documents across sessions. Uploaded files are immediately processed in isolated sandbox directories and purged upon delivery.
- **Hybrid Engine Pipeline:** Pure-PHP libraries (FPDF / FPDI) handle rapid in-memory tasks, while specialized system utilities (QPDF, Ghostscript, Poppler, LibreOffice) handle heavy file manipulation.
- **Dynamic Schema-Driven Registry:** All application tools are declared in a centralized registry (ToolRegistry). The frontend dynamically compiles tool views, inputs, and validation criteria via the REST API.

2. Hardware & Operating System Requirements

The following sizing specifications ensure smooth parallel document operations across concurrent users:

Component	Minimum Requirement	Recommended (Production)	Notes
Operating System	Ubuntu 22.04 LTS / Debian 12	Ubuntu 24.04 LTS	RHEL 9 / Amazon Linux 2023 supported
CPU	1 vCPU (2.0 GHz+)	2 vCPU - 4 vCPU	OCR and conversions scale across cores
System Memory	1 GB RAM	2 GB - 4 GB RAM	LibreOffice and OCRmyPDF require buffer RAM
Disk Space	5 GB Free Storage	20 GB Fast NVMe / SSD	Used for ephemeral request scratchpads
PHP Runtime	PHP 8.3	PHP 8.3+ FPM	cli, fpm, mbstring, xml, zip, intl, gd, curl

3. Core Document Engines & Tool Mapping

PDFolio utilizes dedicated Linux CLI tools tuned for speed and precision:

Engine Binary	System Package	Managed Operations	Key Technical Role
qpdf	qpdf	Split PDF, Rotate PDF	Lossless page extraction, orientation, stripping unreferenced streams
gs (Ghostscript)	ghostscript	Compress PDF	PDFwrite engine: downsamples bitmaps and recompresses streams
pdftoppm	poppler-utils	PDF to Images	Sub-pixel PDF rendering to high-res JPG/PNG raster graphics
pdftotext	poppler-utils	Extract Text	Extracts unicode text streams with column alignment preservation
soffice	libreoffice	Office to PDF	Headless conversion of DOCX, XLSX, PPTX, ODF, CSV to standard PI
ocrmypdf	ocrmypdf + tesseract	OCR PDF	Generates searchable invisible text layer over scanned image pages
FPDI / FPDF	Composer vendor	Merge, Watermark, Numbers	Pure-PHP vector composition, stamping, and pagination headers

4. Server Installation & Configuration

Follow these exact commands to configure an Ubuntu 24.04 LTS host:

Step 1: System Packages & Document Engines

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y php8.3-cli php8.3-fpm php8.3-mbstring php8.3-xml \
    php8.3-zip php8.3-intl php8.3-gd php8.3-curl php8.3-sqlite3 \
    ghostscript qpdf poppler-utils libreoffice \
    ocrmypdf tesseract-ocr tesseract-ocr-osd apparmor-utils
```

Step 2: Ghostscript AppArmor Security Exception

Ubuntu confines Ghostscript via AppArmor, preventing it from accessing files under /var/www. Disable the restriction to allow clean PDF compression:

```
sudo aa-disable /usr/bin/gs
```

Step 3: Web Server & PHP-FPM Upload Tuning

PDFolio accommodates documents up to 100 MB per file (and 220 MB total batch size). Update /etc/php/8.3/fpm/php.ini:

```
upload_max_filesize = 100M
post_max_size       = 220M
max_file_uploads    = 60
memory_limit        = 512M
max_execution_time  = 300
```

In Nginx site configuration (/etc/nginx/sites-available/pdfolio), enforce client upload bounds:

```
client_max_body_size 220M;
```

5. Storage Security & Zero-Retention Lifecycle

Data privacy is enforced by design. When an end-user uploads files, PDFolio creates a cryptographically unique directory in storage/app/pdfolio/{uuid}/. As soon as the deliverable file is generated and streamed back to the browser, PHP's register_shutdown_function triggers an immediate disk purge.

> **IMPORTANT: Automated Pruning Daemon**
> To guard against aborted connections or power interruptions, PDFolio schedules an hourly maintenance worker (php artisan pdfolio:prune) that wipes any orphaned scratch files older than 3 hours.

6. Troubleshooting & System Diagnostics

PDFolio includes an automated self-diagnostic tool to verify your environment:

```
php artisan pdfolio:check
```

Issue Symptom	Root Cause	Definitive Solution
HTTP 413 Payload Too Large	Nginx upload threshold exceeded	Add 'client_max_body_size 220M;' to the Nginx server block
Ghostscript: Unable to open device	Ubuntu AppArmor confinement active	Run 'sudo aa-disable /usr/bin/gs' and restart PHP-FPM
LibreOffice: source file not loaded	Installed via Snap sandbox isolation	Remove Snap ('sudo snap remove libreoffice') and install via APT
Failed to open stream (Permission)	Incorrect ownership on storage folder	Run 'sudo chown -R www-data:www-data storage bootstrap/cache'
Large split PDF file sizes	Legacy parser copying dead objects	Resolved in v2.0: QPDF automatically strips unreferenced streams